

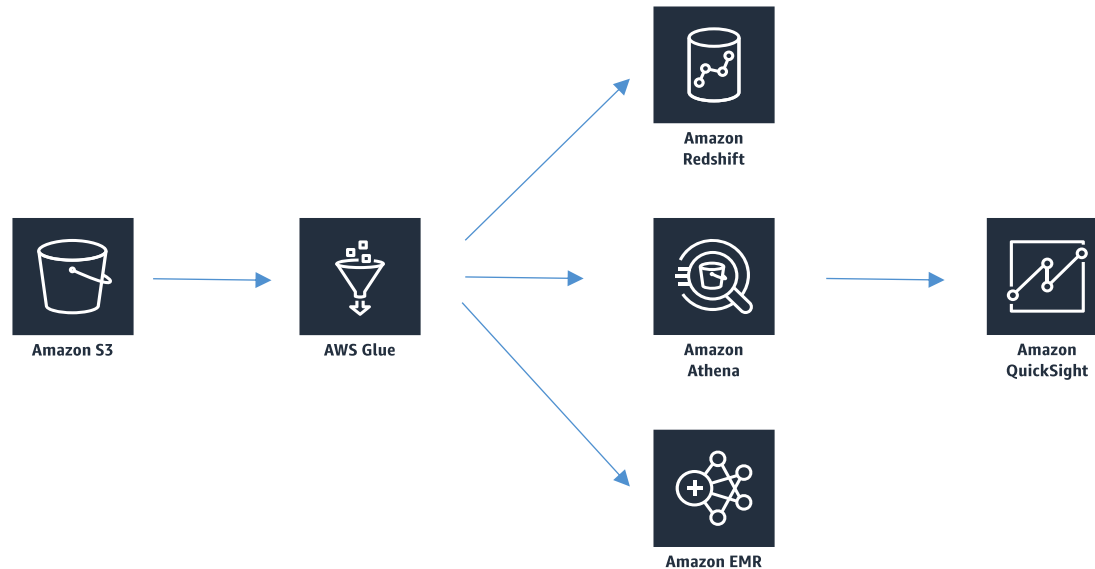
# AWS Glue

Table definitions and ETL

# What is Glue?

- Serverless discovery and definition of table definitions and schema
  - S3 “data lakes”
  - RDS
  - Redshift
  - DynamoDB
  - Most other SQL databases
- Custom ETL jobs
  - Trigger-driven, on a schedule, or on demand
  - Fully managed

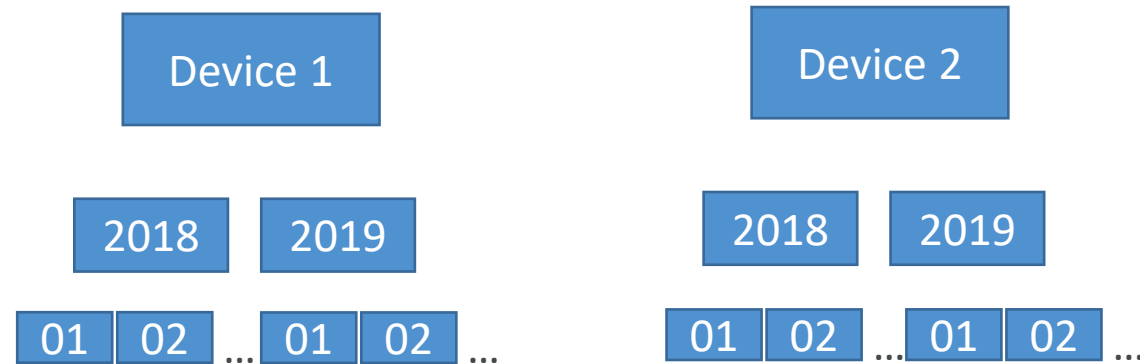
# Glue Crawler / Data Catalog



- Glue crawler scans data in S3, creates schema
- Can run periodically
- Populates the Glue Data Catalog
  - Stores only table definition
  - Original data stays in S3
- Once cataloged, you can treat your unstructured data like it's structured
  - Redshift Spectrum
  - Athena
  - EMR
  - Quicksight

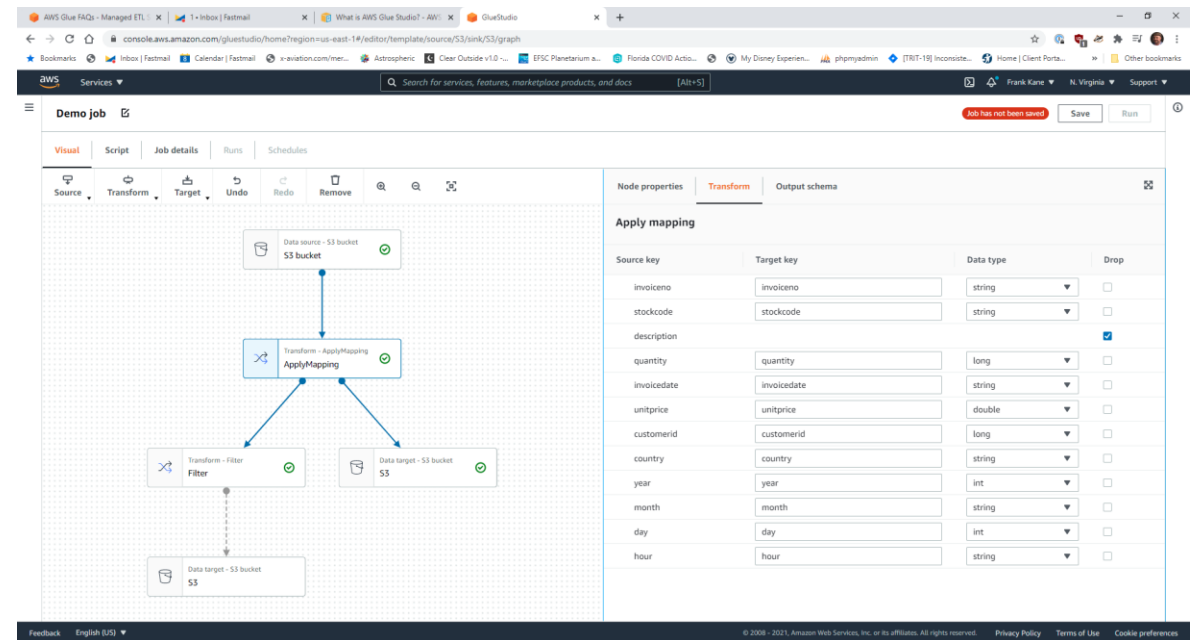
# Glue and S3 Partitions

- Glue crawler will extract partitions based on how your S3 data is organized
- Think up front about how you will be querying your data lake in S3
- Example: devices send sensor data every hour
- Do you query primarily by time ranges?
  - If so, organize your buckets as yyyy/mm/dd/device
- Do you query primarily by device?
  - If so, organize your buckets as device/yyyy/mm/dd



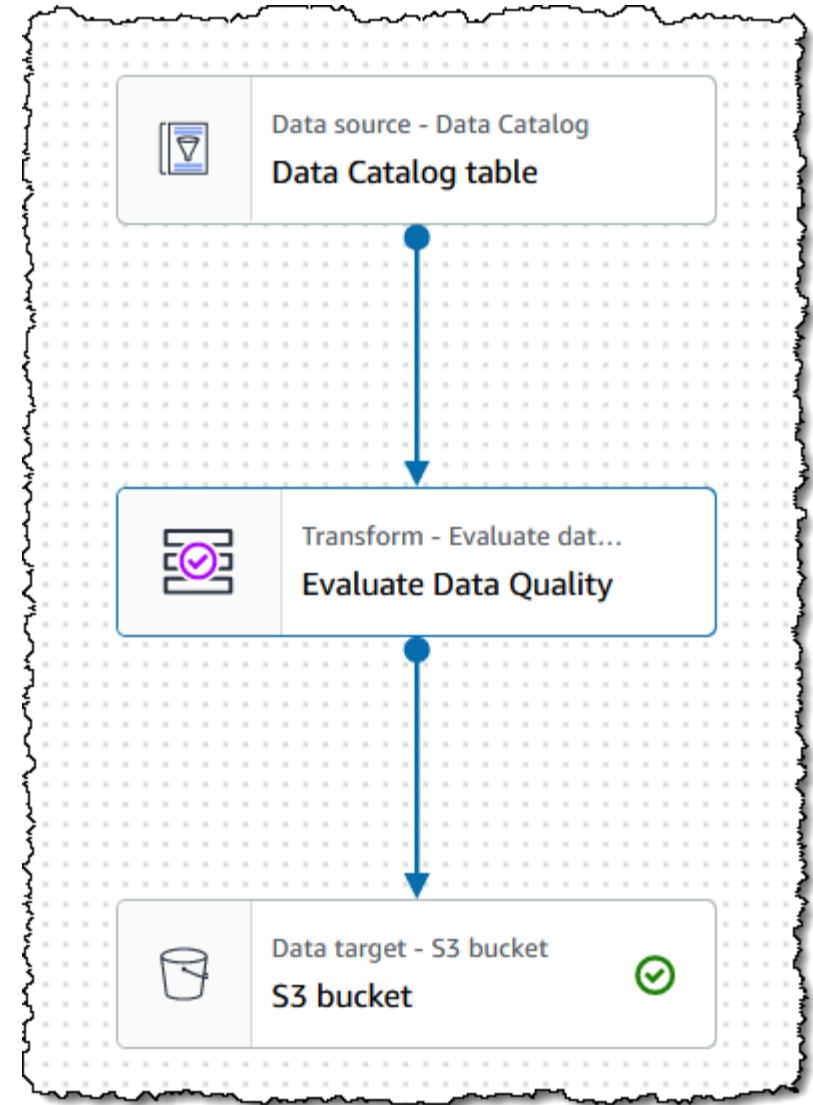
# AWS Glue Studio

- Visual interface for ETL workflows
- Visual job editor
  - Create DAG's for complex workflows
  - Sources include S3, Kinesis, Kafka, JDBC
  - Transform / sample / join data
  - Target to S3 or Glue Data Catalog
  - Support partitioning
- Visual job dashboard
  - Overviews, status, run times



# AWS Glue Data Quality

- Data quality rules may be created manually or recommended automatically
- Integrates into Glue jobs
- Uses Data Quality Definition Language (DQDL)
- Results can be used to fail the job, or just be reported to CloudWatch



# AWS Glue Data Quality

AWS Glue > Tables > tickets > Ruleset details

RS1 Info

⌂

Actions ▾

Evaluate ruleset

Ruleset details

|                                             |                               |         |
|---------------------------------------------|-------------------------------|---------|
| Description                                 | Database                      | Table   |
| Created by Data Quality Rule Recommendation | yyz-tickets                   | tickets |
| Created on                                  | Last modified                 |         |
| -                                           | November 22, 2022 at 17:48:14 |         |

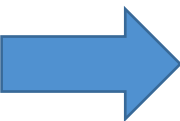
▶ Tags

Rules (DQDL)

Data quality definition language

Copy

Rules = [  
  RowCount between 1021908 and 4087632,  
  IsComplete "TAG\_NUMBER\_MASKED",  
  ColumnLength "TAG\_NUMBER\_MASKED" between 6 and 9,  
  IsComplete "DATE\_OF\_INFRACTION",  
  StandardDeviation "DATE\_OF\_INFRACTION" between 324.19 and 358.31,  
  ColumnValues "DATE\_OF\_INFRACTION" between 20180100 and 20181232,  
  IsComplete "TICKET\_DATE",  
  ColumnLength "TICKET\_DATE" = 10,  
  IsComplete "TICKET\_NUMBER",  
  StandardDeviation "TICKET\_NUMBER" between 16289083634.63 and 18003724017.22,  
  ColumnValues "TICKET\_NUMBER" between 10102501 and 80081081053,  
  IsComplete "OFFICER",  
  StandardDeviation "OFFICER" between 107.99 and 119.35,  
  ColumnValues "OFFICER" <= 394,  
]



Rulesets

Data quality results

dqrun-89769d34ba974a7b9117d6cdf4a26b1108239b38

| Name                                                                                | DQ status   | Status detail                |
|-------------------------------------------------------------------------------------|-------------|------------------------------|
| RS1                                                                                 | DQ failed   | Data quality score is 94.44% |
| Rule_1 RowCount between 1021908 and 4087632                                         | Rule passed |                              |
| Rule_2 IsComplete "TAG_NUMBER_MASKED"                                               | Rule passed |                              |
| Rule_3 ColumnLength "TAG_NUMBER_MASKED" between 6 and 9                             | Rule passed |                              |
| Rule_4 IsComplete "DATE_OF_INFRACTION"                                              | Rule passed |                              |
| Rule_5 StandardDeviation "DATE_OF_INFRACTION" between 324.19 and 358.31             | Rule failed | Expected type of column D    |
| Rule_6 ColumnValues "DATE_OF_INFRACTION" between 20180100 and 20181232              | Rule failed | Expected type of column D    |
| Rule_7 IsComplete "TICKET_DATE"                                                     | Rule passed |                              |
| Rule_8 ColumnLength "TICKET_DATE" = 10                                              | Rule passed |                              |
| Rule_9 IsComplete "TICKET_NUMBER"                                                   | Rule passed |                              |
| Rule_10 StandardDeviation "TICKET_NUMBER" between 16289083634.63 and 18003724017.22 | Rule passed |                              |
| Rule_11 ColumnValues "TICKET_NUMBER" between 10102501 and 80081081053               | Rule passed |                              |
| Rule_12 IsComplete "OFFICER"                                                        | Rule passed |                              |

# AWS Glue DataBrew

- A visual data preparation tool
  - UI for pre-processing large data sets
  - Input from S3, data warehouse, or database
  - Output to S3
- Over 250 ready-made transformations
- You create “recipes” of transformations that can be saved as jobs within a larger project
- May define data quality rules
- May create datasets with custom SQL from Redshift and Snowflake
- Security
  - Can integrate with KMS (with customer master keys only)
  - SSL in transit
  - IAM can restrict who can do what
  - CloudWatch & CloudTrail

```
{  
  "RecipeAction": {  
    "Operation": "NEST_TO_MAP",  
    "Parameters": {  
      "sourceColumns":  
        "[\"age\", \"weight_kg\", \"height_cm\"]",  
      "targetColumn": "columnName",  
      "removeSourceColumns": "true"  
    }  
  }  
}
```

*Example: convert selected columns to key-value pairs*

# Handling Personally Identifiable Information (PII) in DataBrew Transformations

- Enable PII statistics in a DataBrew profile job to identify PII
- Substitution (REPLACE\_WITH\_RANDOM...)
- Shuffling (SHUFFLE\_ROWS)
- Deterministic encryption (DETERMINISTIC\_ENCRYPT)
- Probabilistic encryption (ENCRYPT)
- Decryption (DECRYPT)
- Nulling out or deletion (DELETE)
- Masking out (MASK\_CUSTOM, \_DATE, \_DELIMITER, \_RANGE)
- Hashing (CRYPTOGRAPHIC\_HASH)



# Amazon Athena

Serverless interactive queries of S3 data

# What is Athena?

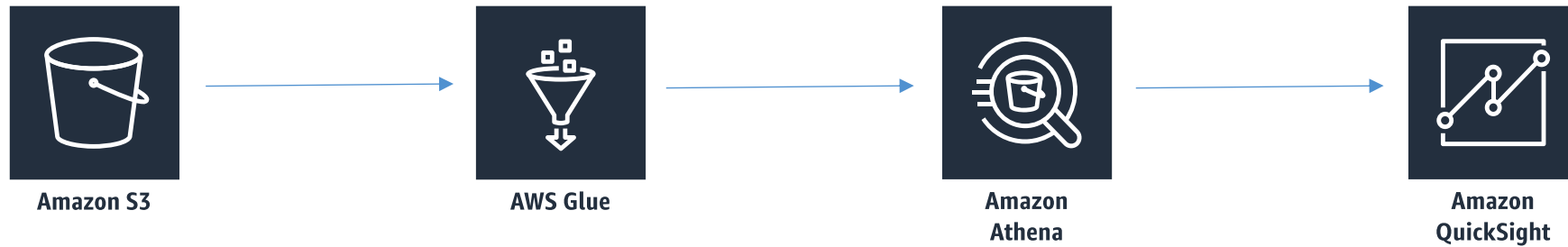
- Interactive query service for S3 (SQL)
  - No need to load data, it stays in S3
- Presto under the hood
- Serverless!
- Supports many data formats
  - CSV, TSV (human readable)
  - JSON (human readable)
  - ORC (columnar, splittable)
  - Parquet (columnar, splittable)
  - Avro (splittable)
  - Snappy, Zlib, LZO, Gzip compression
- Unstructured, semi-structured, or structured

# Some examples

- Ad-hoc queries of web logs
- Querying staging data before loading to Redshift
- Analyze CloudTrail / CloudFront / VPC / ELB etc logs in S3
- Integration with Jupyter, Zeppelin, RStudio notebooks
- Integration with QuickSight
- Integration via ODBC / JDBC with other visualization tools

```
44 GO
45 SELECT p.Name AS ProductName,
46 NonDiscountSales = (OrderQty * UnitPr
47 Discounts = ((OrderQty * UnitPr
48 FROM Production.Product AS p
49 INNER JOIN Sales.SalesOrderData
50 ON p.ProductID = sod.ProductID
51 ORDER BY ProductName DESC;
52 GO
```

# Athena + Glue



# Athena Workgroups

- Can organize users / teams / apps / workloads into Workgroups
- Can control query access and track costs by Workgroup
- Integrates with IAM, CloudWatch, SNS
- Each workgroup can have its own:
  - Query history
  - Data limits (*you can limit how much data queries may scan by workgroup*)
  - IAM policies
  - Encryption settings



# Athena cost model

- Pay-as-you-go
  - \$5 per TB scanned
  - Successful or cancelled queries count, failed queries do not.
  - No charge for DDL (CREATE/ALTER/DROP etc.)
- Save LOTS of money by using columnar formats
  - ORC, Parquet
  - Save 30-90%, and get better performance
- Glue and S3 have their own charges



# Athena Security

- Access control
  - IAM, ACLs, S3 bucket policies
  - AmazonAthenaFullAccess / AWSQuicksightAthenaAccess
- Encrypt results at rest in S3 staging directory
  - Server-side encryption with S3-managed key (SSE-S3)
  - Server-side encryption with KMS key (SSE-KMS)
  - Client-side encryption with KMS key (CSE-KMS)
- Cross-account access in S3 bucket policy possible
- Transport Layer Security (TLS) encrypts in-transit (between Athena and S3)



# Athena anti-patterns

- Highly formatted reports / visualization
  - That's what QuickSight is for
- ETL
  - Use Glue instead



# Athena: Optimizing performance

- Use columnar data (ORC, Parquet)
- Small number of large files performs better than large number of small files
- Use partitions
  - If adding p

E command

